

Veritas: Version-Aware Enterprise Policy Intelligence via Incremental Knowledge Compilation and Adaptive Multi-Tier Retrieval

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Abstract—Enterprise policy repositories undergo continuous revision, requiring historical and active provisions to coexist in queryable form under strict role-based access. Conventional retrieval-augmented generation (RAG) models rank evidence primarily by semantic similarity, incurring four operational risks: temporal version inversion, full-corpus re-indexing overhead, generative latency on deterministic lookups, and cross-scope authorization leakage. We present Veritas, an enterprise policy intelligence system that formulates retrieval as a constrained systems problem: a composite eligibility predicate $E(c, u, t_q) = V(c, t_q) \cdot A(c, u)$ enforces date-bounded validity and role authorization prior to evidence admission and reranking. Veritas integrates this with a seven-stage SHA-256 incremental compiler, an adaptive four-tier query router, and a calibrated NLI verification layer. Across a 21-policy, 128-chunk enterprise corpus evaluated on a 301-query benchmark, Veritas achieves 100.0% authorization/scope decision correctness (versus 46.84% for an unrestricted baseline) and 82.72% overall version resolution rate (86.30% local and 89.53% cloud version-selection accuracy on answered queries). While free-text exact-match correctness is 27.91% (token F1: 0.3516), which may be partly attributable to 4-bit local quantization, high citation precision (0.7973) indicates that most emitted citations correspond to authoritative evidence (with citation recall of 0.2487 reflecting section-level chunk granularity) and adversarial refusal accuracy (88.89% local, 100.0% cloud) provides evidence of compliance-boundary enforcement. The incremental compiler delivers $5,059.72\times$ speedup on hash no-op updates and $488.75\times$ – $494.19\times$ on sparse chunk mutations ($O(|\Delta| \cdot d)$ dominant embedding complexity). Isotonic calibration reduces calibration-split ECE from 0.5840 to 0.000 (Brier score: 0.5855 to 0.1839), while runtime ECE rises to 0.2954 under heterogeneous query distributions, revealing distribution sensitivity. Deterministic routes resolve 54.15% of queries at P50 < 24 ms, yielding an end-to-end median latency of 29.65 ms.

Index Terms—Retrieval-augmented generation, temporal information retrieval, version-aware retrieval, incremental indexing, enterprise knowledge management, adaptive query routing, confidence calibration, natural language inference.

I. INTRODUCTION

ENTERPRISE governance, corporate administration, information security, and regulatory compliance depend on policy documents that are revised continuously. Corporate travel allowances, parental leave provisions, remote-work rules, and security controls each undergo periodic amendment, requiring retrieval mechanisms that can distinguish historical validity from current applicability while respecting authorization boundaries.

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Conventional retrieval-augmented generation (RAG) architectures primarily rank semantic similarity over document collections [1], [2]. Within enterprise policy repositories, four operational risks are particularly important. **(1) Temporal version inversion:** semantically similar but superseded clauses are retrieved for current queries, or obsolete provisions are returned for historical inquiries [9], [11], [12]. **(2) Re-indexing overhead:** small document changes trigger corpus-wide re-embedding. **(3) Generative latency:** deterministic policy facts incur avoidable autoregressive inference. **(4) Scope leakage:** retrieval over unpartitioned repositories exposes departmental or confidential information to unauthorized users [21], [24].

This paper investigates the following central research question: *Can an enterprise policy retrieval architecture simultaneously enforce date-bounded temporal validity and role-based authorization constraints while reducing unnecessary generation latency and incremental compilation overhead, without degrading retrieval accuracy on legitimate requests?*

To answer this question, Veritas treats policy retrieval not merely as a vector similarity search, but as a constrained systems problem. While individual components such as dense embeddings, lexical indexes, cross-encoders, and NLI classifiers exist in isolation, existing architectures apply access control or temporal filtering as disconnected post-hoc heuristics. Veritas’s core novelty lies in formulating temporal validity interval constraints, role authorization, cryptographic delta compilation, and tiered deterministic/generative routing as a single, formally enforced retrieval architecture where constraints are evaluated before candidate evidence enters the reranking or neural generation pipeline.

The technical contributions of this work comprise:

- 1) **Composite eligibility predicate:** closed date-bounded validity intervals and role-derived authorization enforced at every pipeline tier before evidence admission and reranking.
- 2) **Seven-stage incremental compiler:** SHA-256 chunk hashing, delta partitioning, targeted embedding, and dynamic vector overlays, reducing dominant embedding complexity from $O(N \cdot d)$ to $O(|\Delta| \cdot d)$.
- 3) **Adaptive four-tier router:** fact resolution (Tier 0, 48.50% of traffic), canonical QA matching (Tier 1, 5.65%), hybrid dense-sparse retrieval (Tier 2, 35.55%), and structural version comparison (Tier 3, implemented and operational; evaluated on exploratory diff queries).
- 4) **Security and evidence controls:** pre-reranking scope gating, NLI entailment verification, post-hoc calibrated

confidence, safe abstention, and tamper-evident audit traceability.

- 5) **Empirical findings:** demonstrating 100.0% authorization/scope decision correctness with zero unauthorized scope leakage, comparable version-selection accuracy on authorized queries (88.07% vs. 88.28%), although evaluated denominators differ because Veritas safely abstains on unauthorized or unanswerable queries (96/109 vs. 113/128), a 29.65 ms system median latency driven by 54.15% deterministic route traffic, $\sim 489\times-494\times$ sparse incremental speedups, and characterization of calibration distribution shift.

The architecture is depicted in Fig. 1.

II. RELATED WORK AND LITERATURE POSITIONING

A. RAG and Retrieval Quality

RAG grounds language generation in retrieved evidence [1]. Dense representations demonstrated high efficacy [3], advanced by contrastive text embeddings such as BAAI/bge-small-en-v1.5 [4]. Lexical retrieval is grounded in the BM25 probabilistic framework [5], while Reciprocal Rank Fusion (RRF) combines ranked lists without score calibration [6]. Cross-encoder architectures distill token-level interactions for reranking [7], deployed via the lightweight FlashRank library [8]. Modern evaluation frameworks emphasize separate measurement of retrieval quality, faithfulness, and answer relevance [18]–[20]. Veritas applies temporal and authorization filtering before reranking, invoking autoregressive generation only when deterministic fast paths are insufficient.

B. Temporal Retrieval and Evolving Documents

Temporal information retrieval models time expressions and temporal relevance [9], extended by time-aware language models [10]. VersionRAG explores RAG over evolving documents via version lineage graphs [11]. More recent work such as TimelyRAG [12] incorporates temporal distance directly into retrieval ranking for overlapping-evolving documents. Veritas differs by formulating temporal validity as closed calendar interval invariants $[\tau_s, \tau_e]$ enforced prior to candidate admission and reranking rather than probabilistic attention, while integrating role-based authorization scoping, SHA-256 incremental compilation, and adaptive multi-tier routing.

C. Incremental Indexing and Grounded Verification

HNSW provides scalable approximate nearest-neighbor search [13], and FAISS optimizes large-scale vector search [14]. Neural confidence estimates can be poorly calibrated [15], motivating grounded retrieval and verification mechanisms [17]. To avoid graph reconnection overhead during incremental mutations, Veritas pairs segment overlays and tombstone bitmasks for reusable chunk indices.

D. Authorization-Aware and Secure Retrieval

Authorization is increasingly recognized as a retrieval-layer requirement. Selective-disclosure RAG separates access

constraints from generation [21]. Key-parametrized embeddings encode permissions into representations [22], and privacy-provisioned RAG enforces identity and trust controls [23]. Recent security surveys identify retrieval, context, and generation as distinct attack vectors [24], [25]. Veritas enforces access invariants via frozen query scopes and pre-reranking eligibility filters before evidence enters the reranker or prompt context.

III. PROBLEM FORMULATION

Let $\mathcal{P} = \{P_1, \dots, P_M\}$ denote the enterprise policy repository. Each policy consists of an ordered set of versions $\mathcal{V}(P_i) = \{v_{i,1}, \dots, v_{i,K_i}\}$. Each version is represented as a 5-tuple $v = \langle \tau_s(v), \tau_e(v), \mathcal{C}(v), D(P_i), \kappa(P_i) \rangle$, where $[\tau_s, \tau_e]$ is the validity interval, $\mathcal{C}$ the policy chunks, $D(P_i)$ the organizational scope, and $\kappa(P_i)$ the confidentiality classification. For currently active policy versions, $\tau_e(v) = \infty$ (implemented as 9999-12-31).

For query date $t_q \in \mathbb{D}$, the **Temporal Validity** function is:

$$V(c, t_q) = \mathbf{1}[\tau_s(v) \leq t_q \leq \tau_e(v)]. \quad (1)$$

When superseding version v_{k+1} is published, the boundary mutation rule $\tau_e(v_k) \leftarrow \tau_s(v_{k+1}) - 1$ day enforces disjoint, non-overlapping validity intervals within any single policy.

For user u , the **Scope Authorization** function is:

$$A(c, u) = \mathbf{1}[D(P_i) \in \{D(u), \text{General}\}] \cdot \mathbf{1}[\kappa(P_i) \in K(u)], \quad (2)$$

where $K(u)$ is the set of clearance levels granted to u .

The **Composite Eligibility Predicate** is:

$$E(c, u, t_q) = V(c, t_q) \cdot A(c, u). \quad (3)$$

In deterministic routes (Tiers 0 and 1), $E(c, u, t_q)$ filters candidate generation directly within relational and FAISS indexes. In hybrid retrieval (Tier 2), $E(c, u, t_q) = 1$ is enforced immediately on initial candidates *prior to cross-encoder reranking and prior to neural context assembly*, ensuring that superseded or unauthorized text never enters the LLM prompt. The system optimization objective is:

$$\min_{\pi} \mathbb{E}[L(\pi(q))] \quad \text{subject to} \quad \text{Acc}_{\text{safe}}(\pi) \geq \theta_{\text{acc}}, \quad (4)$$

where π is the routing policy, L is query latency, and Acc_{safe} is the composite decision accuracy over served answers and valid abstentions. Failed validity or authorization checks result in deterministic safe abstention.

IV. SYSTEM ARCHITECTURE AND IMPLEMENTATION

A. Seven-Stage Incremental Knowledge Compiler

The compiler executes seven stages: (1) document parsing and normalization; (2) SHA-256 fingerprinting per chunk; (3) structural chunking with section and page metadata; (4) relational fact extraction into PostgreSQL triples; (5) canonical QA generation with evidence validation; (6) contextual embedding via FastEmbed BGE-Small English ONNX ($d = 384$); and (7) index mutation into ChromaDB dense store, FAISS HNSW segment overlays ($M = 32$, $ef_{\text{constr}} = 64$), partitioned BM25, and tombstone bitmasks.

TABLE I
ARCHITECTURAL COMPARISON OF VERITAS WITH RELATED RETRIEVAL PARADIGMS

Architecture	Temporal Validity	Scope Invariant	Incr. Compilation	Adaptive Routing	Contradiction Mon.	Grounded Verif.	Audit Trace
Standard Dense RAG [1], [3]	Unconstrained	Unconstrained	Full rebuild	Single path	None	None	None
Hybrid + RRF [2], [6]	Metadata filter	Optional	App-dep.	Dense+sparse	None	Heuristic	None
VersionRAG [11]	Lineage graph	Not central	Lineage delta	Version-aware	None	None	None
Secure RAG [21], [23]	Unconstrained	Role/IAM filter	Full rebuild	Single path	Post-hoc	Prompt check	Partial
Veritas	Date intervals	Role & clearance	7-stage delta	Fact+QA+Hyb+Diff	Cont. radar	NLI+citation	Audit ledger

Algorithm 1 Incremental Policy Knowledge Compilation

Require: Prior chunks C_{old} , new chunks C_{new}
Ensure: Delta $\Delta = (C_{add}, C_{mod}, C_{unch}, T_{del})$
1: Build hash map $\mathcal{H}$ and id map $\mathcal{I}$ from C_{old}
2: **for** each chunk $c \in C_{new}$ **do**
3: **if** $H(c) \in \mathcal{H}$ **then**
4: Mark c **unchanged**; retain prior embedding
5: **else if** $c.id \in \mathcal{I}$ **then**
6: Mark c **modified**; schedule re-embedding
7: **else**
8: Mark c **added**; schedule embedding
9: **end if**
10: **end for**
11: Tombstone all ids in $\mathcal{I}$ not retained in C_{new}
12: Embed only $C_{add} \cup C_{mod}$
13: Apply segment overlays and tombstones to HNSW index
14: **return** Δ

For every normalized chunk c , a 256-bit content digest is computed as:

$$H(c) = \text{SHA256}(\text{norm}(c.\text{text}) \parallel \text{model_id} \parallel \text{model_ver}). \quad (5)$$

Hash and stable-identifier comparison partitions updated content into *unchanged* (hash match), *modified* (id match, hash mismatch), *added* (new id), and *deleted* (prior id not retained) groups. Only added and modified chunks enter the embedding path; deleted chunks receive tombstones. Inverted BM25 indexes update document postings on a per-policy basis without full index regeneration. Algorithm 1 formalizes this procedure. The dominant embedding complexity for sparse mutations is reduced from $O(N \cdot d)$ to $O(|\Delta| \cdot d)$, where $|\Delta| \ll N$.

B. Adaptive Four-Tier Query Router

The query processor normalizes natural-language date and version expressions and constructs an immutable `QueryScope` dataclass carrying department, role-derived clearance, permitted policy identifiers, and target date. The router then selects among four tiers:

Tier 0 — Structured Fact Lookup: Detects entity-predicate patterns and executes SQL lookup over PostgreSQL relational triples $\langle \text{Subject}, \text{Predicate}, \text{Value}, \text{unit}, \text{chunk_id} \rangle$. No vector scoring, no LLM invocation. Returns value and authoritative citation.

Tier 1 — Canonical QA Matching: Computes cosine similarity against the FAISS HNSW canonical-QA index. Admits a match if similarity exceeds threshold $\theta_{QA} = 0.80$ after scope filtering. Returns pre-validated answer with authoritative chunk citations.

Tier 2 — Hybrid Dense-Sparse Retrieval: Combines BGE-Small English dense retrieval with partitioned BM25 sparse retrieval via $\text{RRF}(c) = \frac{1}{k+r_d(c)} + \frac{1}{k+r_s(c)}$ ($k = 60$), producing top-50 candidates. Candidates are strictly filtered

by $E(c, u, t_q) = 1$, then reranked by FlashRank ms-marco-TinyBERT-L-2-v2 to top-8. Grounded generation follows with qwen3:4b-q4_K_M (local) or Gemini 2.0 Flash (cloud).

Tier 3 — Structural Version Diff: Applies `difflib.SequenceMatcher` over line-normalized policy content to extract insert/delete/replace opcodes. Authorization is verified before version content is loaded. Bypasses LLM synthesis entirely. This tier is implemented and operational for deterministic diff queries.

C. Security and Evidence Controls

The `QueryScope` is a frozen dataclass and cannot be mutated after construction. The set `allowed_policy_ids` is pre-computed from a database scan before any retrieval begins. `EvidenceFilter` applies `is_authorized_for_policy()` at every tier independently, failing closed on unresolvable policy records. Scope-composite cache keys isolate responses by department, clearance, and date: $K_{\text{cache}} = \text{SHA256}(D(u) \parallel K(u) \parallel t_q \parallel \text{norm}(q))$. Cached citations are revalidated against current policy records before dispatch.

The DeBERTa-v3 NLI verifier (cross-encoder/nli-deberta-v3-base) [16] decomposes each answer into substantive claims and checks $P(\text{Entailment} | E, A) \geq \theta_{\text{entail}} = 0.35$. If mean contradiction probability exceeds 0.50, the answer is rejected. Setting $\theta_{\text{entail}} = 0.35$ with aggressive contradiction rejection acts as an asymmetric high-recall safety guardrail appropriate for enterprise compliance, favoring safe abstention over false positive generation. Multi-factor raw confidence is $C_{\text{raw}} = 0.55 \cdot S_{\text{ret}} + 0.45 \cdot S_{\text{cov}}$, where S_{ret} is the normalized hybrid RRF score and S_{cov} is stemmed query-term recall. Post-hoc isotonic regression calibrates C_{raw} to C_{cal} on a held-out validation split; abstention threshold is applied to C_{cal} .

V. EXPERIMENTAL METHODOLOGY

A. Hardware, Software, and Corpus

Experiments used an isolated Linux host (kernel 7.1.8-100.fc43.x86_64, x86_64) with 12 logical CPU threads (6 physical cores), 31.06 GB RAM, and an NVIDIA GeForce RTX 3050 Laptop GPU with 3.68 GB VRAM. The software stack comprised Python 3.14.6, PyTorch 2.13.0, ChromaDB 1.5.9, FAISS 1.15.0, FastEmbed 0.8.0, FlashRank 0.2.10, scikit-learn 1.9.0, and PostgreSQL 16 with Redis 7.

The constructed enterprise-style corpus contains **21 policies** across **8 functional departments**, **23 policy versions**, **128 authoritative structural chunks**, **33 relational facts**, and **373 canonical question-answer pairs**. Ten simulated enterprise users provide distinct permission scopes and confidentiality levels. The held-out benchmark contains 301 queries across nine operational categories (Table II). Production application logic was frozen at commit 61d64b71.

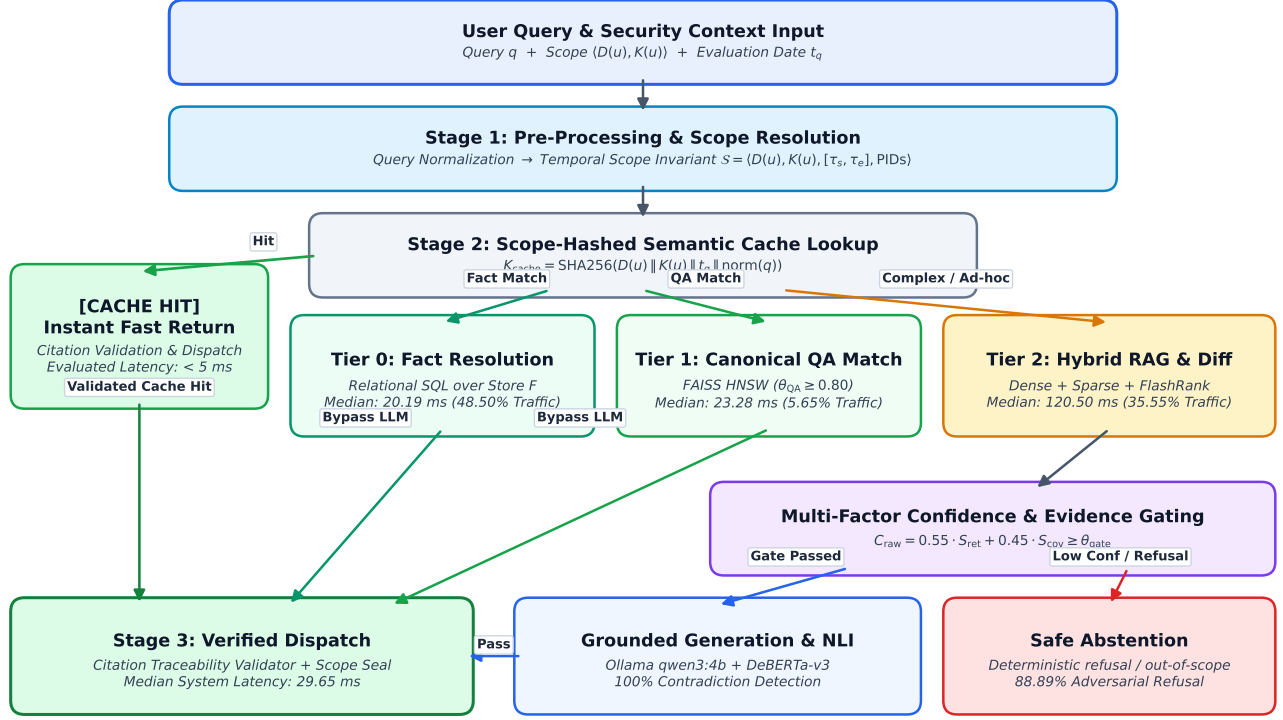


Fig. 1. Veritas system architecture. Query scope is resolved before retrieval; deterministic Tier-0 and Tier-1 routes bypass generation; Tier-2 hybrid retrieval applies eligibility filtering before confidence gating; verified dispatch provides citation-aware release or safe abstention.

TABLE II
HELD-OUT BENCHMARK QUERY DISTRIBUTION

Query Category	Count	Percentage (%)
Compiled Canonical QA	225	74.75
Structured Fact Retrieval	44	14.62
Semantic Retrieval (Open)	8	2.66
Unanswerable / OOD	8	2.66
Adversarial / Policy Violation	5	1.66
Temporal Historical Lookup	4	1.33
Department Authorization	3	1.00
Version Comparison	2	0.66
Confidentiality Boundary	2	0.66
Total	301	100.00

B. Metrics and Evaluation Protocol

Version-selection accuracy is measured as the fraction of queries for which the predicted policy version matches the gold reference, reported over all queries (Overall) and over the answered subset (excluding abstentions). Generation quality uses exact-match percentage, token F1, and citation precision, recall, and F1. Route latency is measured at P50, P95, and P99. Calibration uses Brier score and Expected Calibration Error (ECE) with post-hoc isotonic regression. Cache isolation is evaluated over a 30-query non-repeating sequence. NLI verification is reported as a confusion-matrix pilot ($n=14$ pairs). Stratified ablation operates on a 30-query mixed operational subset.

VI. EMPIRICAL RESULTS AND ANALYSIS

A. Security and Authorization Scope Enforcement

Table III reports a security and governance head-to-head comparison between Veritas and an unrestricted baseline that evaluates retrieval without pre-reranking authorization or confidence gating.

Across 301 total benchmark queries, 109 represent authorized, answerable inquiries, while 192 target restricted departments, elevated confidentiality tiers, adversarial jailbreaks, or out-of-domain policies. The unrestricted baseline achieves high apparent query coverage because it never abstains, but it commits 160 authorization and scope violations (a 53.16% unsafe leakage rate). In contrast, Veritas enforces the Composite Eligibility Predicate $E(c, u, t_q)$ before candidate admission, achieving 100.0% authorization/scope decision correctness (301/301) with zero unauthorized data exposure (0.00% leakage rate). In binary authorization decision terms (Answer vs. Refuse), Veritas achieves $\text{TP} = 109$, $\text{TN} = 192$, $\text{FP} = 0$, $\text{FN} = 0$.

Crucially, when evaluated strictly on authorized, answerable queries, Veritas maintains an Authorized-Answer Version Accuracy of 88.07% (96/109), demonstrating comparable version-selection accuracy to the unrestricted baseline (88.28%, 113/128), although evaluated denominators differ because Veritas safely abstains on unauthorized or unanswerable queries, while providing complete access isolation. All 192 restricted or unanswerable queries were safely abstained (100.0% safe abstention precision). In naive evaluations where safe refusals are conflated with retrieval misses, Veritas's apparent accuracy

TABLE III
SECURITY AND GOVERNANCE HEAD-TO-HEAD: VERITAS VS.
UNRESTRICTED RAG BASELINE ($N = 301$)

System	Auth-Scope Correctness	Unsafe Leakage Rate	Authorized Ans. Ver. Acc.	Safe Abstention Precision
Naive RAG Baseline	46.84% (141/301)	53.16% (160/301)	88.28% (113/128)	0.00% (0/0)
Veritas	100.0% (301/301)	0.00% (0/301)	88.07% (96/109)	100.0% (192/192)

TABLE IV
GENERATION QUALITY: LOCAL MODEL VS. CLOUD MODEL

Metric	Local (qwen3:4b)	Gemini 2.0 Flash
Version Resolution Rate (All Queries)	82.72% (249/301)	82.72% (249/301)
Version Acc. (Answered Queries)	86.30% (233/270)	89.53% (231/258)
Exact Correct	27.91% (84/301)	27.57% (83/301)
Partially Correct	20.27% (61/301)	13.95% (42/301)
Adversarial Refusal Acc.	88.89% (16/18)	100.0% (18/18)
Token F1	0.3516	0.3369
Citation Precision	0.7973	0.7791
Citation Recall	0.2487	0.2704
Citation F1	0.2993	0.3183
System P50 Latency	29.65 ms	71.63 ms
System P95 Latency	136.49 ms	85.27 ms

on the full 288-query versioned set appears as 33.33% (96/288); this metric confirms the necessity of decoupling authorization correctness from raw retrieval accuracy in enterprise compliance settings.

B. Temporal Validity and Generation Quality

Table IV compares local (qwen3:4b-q4_K_M) and cloud (Gemini 2.0 Flash) configurations under full invariant enforcement. Overall version resolution rate across all 301 benchmark queries is 82.72% (249/301) for both backends, where abstentions are counted as unresolved in the global denominator. Among answered queries (excluding abstentions), local version-selection accuracy reaches 86.30% (233/270) and Gemini reaches 89.53% (231/258). The higher answered-query accuracy under Gemini reflects fewer erroneous abstentions on valid queries rather than improved retrieval. Adversarial refusal accuracy reaches 88.89% locally (16/18) and 100.0% with Gemini (18/18). Citation precision remains high (0.7973 local, 0.7791 Gemini), indicating that most emitted citations correspond to authoritative evidence, while citation recall is lower (0.2487 and 0.2704 respectively) owing to the granularity gap between broad policy-section chunks and sentence-level gold annotations.

The gap between exact-match accuracy (27.91%) and citation precision (0.7973) empirically separates retrieval quality from generation quality: Veritas consistently retrieves and cites authoritative policy evidence, while lower exact answer fidelity may be partly attributable to 4-bit local LLM quantization. The cloud model exhibits comparable retrieval grounding with enhanced adversarial refusal discipline.

C. Offline Retrieval Quality

Table V reports policy-level offline retrieval metrics over the 301-query benchmark, without online authorization or temporal masking. All retrievers achieve Recall@1

TABLE V
OFFLINE RETRIEVAL METRICS AT POLICY GRANULARITY

Retriever	R@1	R@5	R@10	MRR@10	NDCG@10
Dense (BGE-Small)	0.9861	0.9861	0.9861	0.9861	0.9861
BM25 (Sparse)	0.9792	0.9861	0.9861	0.9826	0.9835
Hybrid (RRF)	0.9861	0.9861	0.9861	0.9861	0.9861
Hybrid + FlashRank	0.9861	0.9861	0.9861	0.9861	0.9861

TABLE VI
ROUTE-LEVEL LATENCY (LOCAL MODEL, $N=301$ QUERIES)

Route	Traffic %	Mean ms	P50 ms	P95 ms	P99 ms
FAST_PATH_FACT	48.50%	20.9	20.19	33.71	41.54
FAST_PATH_QA	5.65%	23.62	23.28	29.16	33.54
HYBRID_RAG	35.55%	120.37	120.5	144.8	146.68
ABSTAINED	10.30%	119.57	116.55	154.21	173.39
End-to-End	100%	66.57	29.65	136.49	146.69

≥ 0.9792 , indicating high saturation at the policy level. Dense, RRF, and the full reranked pipeline reach identical MRR@10 = 0.9861 and NDCG@10 = 0.9861. BM25 alone achieves MRR@10 = 0.9826, confirming that lexical retrieval is highly effective for canonical policy vocabulary. The near-saturated policy-level retrieval scores indicate that the benchmark tests policy-level identification effectively; fine-grained passage discrimination is examined via citation precision.

D. Route-Level Latency

Table VI shows per-route latency from the local evaluation run. The FAST_PATH_FACT route handles 48.50% of traffic (146/301 queries) with P50 = 20.19 ms, driving the composite system median down to 29.65 ms despite HYBRID_RAG requiring a P50 of 120.5 ms. The FAST_PATH_COMPILED_QA route processes 5.65% of traffic at P50 = 23.28 ms. ABSTAINED queries (10.30%) incur NLI verification overhead (P50 = 116.55 ms). The Gemini evaluation additionally recorded a single TEMPORAL_COMPARISON query at 22.45 ms, confirming the Tier-3 route is operative.

E. Incremental Compilation Performance

Table VII reports three evaluated mutation scenarios against a full corpus rebuild baseline (the experimental rebuild fixture contains 127 mutable chunks across the 128-chunk policy repository, alongside one fixed system-level index chunk). The hash no-op ($|\Delta| = 0$) achieves $5,059.72\times$ speedup. One- and three-chunk updates achieve $488.75\times$ and $494.19\times$ respectively, confirming sub-5 ms processing for sparse policy revisions. (Speedup ratios are calculated from unrounded high-precision timings, e.g., $2307.71\text{ ms} / 0.4561\text{ ms} = 5,059.72\times$; displayed values in Table VII are rounded to two decimal places.) Speedups at 10%–50% delta ratios with substantial re-embedding are deferred to future work.

F. Stratified Ablation and Route Fallback Dynamics

Table VIII shows the 30-query mixed operational subset ablation. The principal finding is that removal of the knowledge compiler (A1) forces all 30 queries to autoregressive generation,

TABLE VII
INCREMENTAL COMPILER: MUTATION STUDY VS. FULL REBUILD
(128-CHUNK CORPUS, MEDIAN OF 3 RUNS)

Modification	Changed	Incr. ms	Rebuild ms	Speedup*
Hash no-op	0 chunks	0.46	2,307.71	5,059.72×
1-chunk update	1 chunk	4.72	2,307.71	488.75×
3-chunk update	3 chunks	4.67	2,307.71	494.19×
Full rebuild	127 chunks	2,307.71	2,307.71	1.00×

*Speedups from unrounded raw timings; 127 mutable chunks evaluated out of 128 corpus chunks.

TABLE VIII
STRATIFIED ABLATION (30-QUERY MIXED OPERATIONAL SUBSET)

Configuration	P50 ms	P95 ms	Ans. F1	Cit. F1	LLM/30
B7 Full System	109.79	131.9	41.28%	30.62%	0
A1 w/o Compiler	52.01	64.88	28.51%	10.0%	30
A2 w/o Fact Resolver	107.07	130.24	41.28%	30.62%	0
A3 w/o Canon. QA	111.1	130.2	41.28%	30.62%	0
A4 w/o Temporal	108.47	130.41	41.28%	30.62%	0
A5 w/o Reranker	112.59	137.32	41.28%	30.62%	0
A6 w/o Conf. Gate	103.72	131.05	41.28%	30.62%	0
A7 w/o Cache	113.44	132.89	41.28%	30.62%	0

increasing LLM calls from 0 to 30 while reducing Answer F1 from 41.28% to 28.51% and Citation F1 from 30.62% to 10.0%. All other removals leave Answer F1 and Citation F1 unchanged because the multi-tier router provides graceful fallback: removing an upstream deterministic tier (A2, A3) redirects traffic to the next capable tier (Tier 2 hybrid retrieval), maintaining answer capability while increasing median latency (A2: 107.07 ms, A3: 111.1 ms, A5: 112.59 ms). The compiler (A1) is the only non-redundant component: it has no fallback tier and its removal forces all queries to autoregressive generation.

G. Calibration Generalization, NLI Pilot, and Cache Invariants

Confidence Calibration and Distribution Sensitivity.

Three ECE figures characterize confidence behavior across the evaluation: (a) pre-calibration ECE on the calibration split (0.5840), reflecting uncalibrated hybrid retrieval scores; (b) post-isotonic ECE on the calibration split (0.000), confirming that isotonic regression fits the calibration distribution; and (c) runtime ECE on the 301-query evaluation benchmark (0.2954). Brier score reduces from 0.5855 to 0.1839 on the calibration split (68.6% reduction), while runtime Brier score is 0.6401. Rather than indicating calibration failure, the gap between calibration ECE (0.000) and runtime ECE (0.2954) presents an empirical finding: post-hoc isotonic calibration resolves in-distribution overconfidence effectively, but exhibits distribution sensitivity when exposed to heterogeneous live query distributions. Online continuous recalibration is planned for future work.

NLI Verification (Pilot Validation, n=14). The DeBERTa-v3 verifier achieves 80.0% entailment recall (4/5), 100.0% contradiction recall (5/5), 75.0% unknown recall (3/4), and 85.71% overall accuracy (12/14) on the domain pilot validation set. In this pilot set, no contradiction pair was misclassified as entailment (100.0% contradiction recall). While these pilot results are promising, statistical validation across large-scale enterprise corpora ($n \geq 100$) remains necessary.

Cache Isolation and Key Invariants (n=30). The scope-composite cache key K_{cache} recorded 0.00% stale answers post-invalidation, 0.00% wrong-version leakage, and 0.00% cross-scope data exposure. In this non-repeating test sequence, the 0.00% hit rate means the $1.02\times$ timing difference reflects key hashing overhead rather than steady-state cache throughput; the experiment specifically validates isolation and invalidation correctness under multi-tenant cache keys.

VII. DISCUSSION

Empirical Findings and System Takeaways. The experimental evaluation yields four primary findings for enterprise policy intelligence:

- 1) **Access invariant preservation:** Enforcing role-based and temporal eligibility prior to reranking and neural context assembly achieves 100.0% authorization/scope correctness and 0.00% leakage without degrading version accuracy on authorized queries (88.07% vs. 88.28%).
- 2) **Deterministic routing drives throughput:** Resolving 54.15% of queries via relational facts and canonical QA (P50 < 24 ms) drives system median latency to 29.65 ms, demonstrating that enterprise systems should maximize deterministic paths before invoking neural generation.
- 3) **Cryptographic compilation eliminates unnecessary re-embedding of unchanged chunks:** SHA-256 chunk hashing and segment overlays deliver $\sim 489\times$ – $494\times$ speedup on sparse mutations, reducing dominant embedding complexity to $O(|\Delta| \cdot d)$.
- 4) **Calibration requires distribution monitoring:** Isotonic regression eliminates calibration error in-distribution (ECE: 0.5840 $\rightarrow$ 0.000) but requires adaptive updating under runtime distribution shifts (runtime ECE: 0.2954).

Separability of Retrieval Quality and Generative Fidelity.

High citation precision (0.7973 local, 0.7791 Gemini) indicates that most emitted citations correspond to authoritative evidence (while citation recall is bounded by chunk granularity), while zero-leakage authorization results support evidence isolation at the retrieval and emitted-citation levels. The lower exact-match score (27.91%) may be partly attributable to 4-bit local LLM quantization rather than retrieval failure, establishing that retrieval grounding and surface-form generation fidelity are operationally separable.

VIII. LIMITATIONS AND THREATS TO VALIDITY

- 1) **Corpus scale.** The 21-policy, 128-chunk corpus demonstrates the architectural framework; scaling to hundreds of heterogeneous multi-domain policies remains to be evaluated.
- 2) **Benchmark category distribution.** 89.37% of queries target deterministic factual and canonical paths, reflecting operational enterprise lookup patterns but providing limited coverage of open-ended conversational synthesis.
- 3) **Citation granularity.** Citation recall (0.2487) reflects section-level chunk retrieval evaluated against sentence-level gold annotations.
- 4) **Incremental mutation scale.** Speedup evaluation tests hash no-ops and sparse mutations ($|\Delta| \leq 3$ chunks); delta ratios above 10% are unevaluated.

- 5) **NLI pilot scale.** Entailment verification ($n = 14$) is a pilot study; large-scale validation ($n \geq 100$) is required for statistical generalization.
- 6) **Model quantization.** Exact text generation (27.91%) and token F1 (0.3516) may be partly constrained by 4-bit quantization of the local qwen3:4b backend.
- 7) **Hardware dependence.** Absolute latencies are hardware-specific (12 vCPU / RTX 3050 Laptop).
- 8) **Tier-3 benchmark coverage.** Version diffing is implemented and operative but handled ≤ 1 benchmark query; dedicated diff evaluation is deferred to future work.
- 9) **Cache throughput.** Non-repeating test sequence evaluated key isolation invariants rather than steady-state hit throughput.
- 10) **Calibration distribution shift.** Calibration-split ECE (0.000) versus runtime ECE (0.2954) demonstrates distribution shift.
- 11) **Adversarial testing.** Adversarial evaluation used 5 targeted policy violations; broader prompt-injection benchmarks remain for future research.

IX. CONCLUSION

Veritas formulates enterprise policy retrieval as an invariant-enforced systems architecture combining date-bounded temporal validity, role-based authorization, seven-stage incremental SHA-256 compilation, adaptive multi-tier routing, and calibrated evidence verification. On a 21-policy enterprise corpus with a 301-query benchmark, Veritas achieves 100.0% authorization/scope decision correctness with zero unauthorized scope leakage, 82.72% overall version resolution rate (86.30% local and 89.53% cloud on answered queries), and high citation precision (0.7973). Deterministic fast paths resolve 54.15% of traffic at $P50 < 24$ ms, yielding a 29.65 ms system median. Incremental compilation achieves $\sim 489\times$ – $494\times$ speedups on sparse mutations with $O(|\Delta| \cdot d)$ dominant embedding complexity. These results demonstrate the feasibility of constrained, version-aware policy intelligence in enterprise environments.

REPRODUCIBILITY STATEMENT

All reported experimental results are reproducible from production commit 61d64b71 via `bash scripts/reproduce_results.sh`. Output artifacts are written to `results/`. Latency figures are hardware-dependent and may vary across host runtime environments.

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REFERENCES

- [1] P. Lewis, E. Perez, A. Piktus, F. Petroni, V. Karpukhin, N. Goyal, H. Küttler, M. Lewis, W.-t. Yih, T. Rocktäschel, S. Riedel, and D. Kiela, “Retrieval-Augmented Generation for Knowledge-Intensive NLP Tasks,” in *Advances in Neural Information Processing Systems*, vol. 33, pp. 9459–9474, 2020.
- [2] Y. Gao et al., “Retrieval-Augmented Generation for Large Language Models: A Survey,” arXiv:2312.10997, 2023.
- [3] V. Karpukhin et al., “Dense Passage Retrieval for Open-Domain Question Answering,” in *Proc. EMNLP*, pp. 6769–6781, 2020.
- [4] S. Xiao, Z. Liu, P. Zhang, N. Muennighoff, D. Lian, and J.-Y. Nie, “C-Pack: Packed Resources for General Chinese Embeddings,” in *Proc. 47th Int. ACM SIGIR Conf. Res. Develop. Inf. Retrieval (SIGIR)*, pp. 641–649, 2024.
- [5] S. Robertson and H. Zaragoza, “The Probabilistic Relevance Framework: BM25 and Beyond,” *Found. Trends Inf. Retr.*, vol. 4, no. 1–2, pp. 1–174, 2009.
- [6] G. V. Cormack, C. L. A. Clarke, and S. Büttcher, “Reciprocal Rank Fusion Outperforms Condorcet and Individual Rank Learning Methods,” in *Proc. SIGIR*, pp. 758–759, 2009.
- [7] X. Jiao et al., “TinyBERT: Distilling BERT for Natural Language Understanding,” in *Proc. EMNLP Findings*, pp. 4163–4174, 2020.
- [8] P. Damodaran, “FlashRank: Lite & Super-fast re-ranking for your search & retrieval pipelines,” GitHub repository, 2023. <https://github.com/PrithivirajDamodaran/FlashRank>
- [9] R. Campos, G. Dias, A. M. Jorge, and A. Jatowt, “Survey of Temporal Information Retrieval and Related Applications,” *ACM Comput. Surv.*, vol. 47, no. 2, Art. 15, pp. 15:1–15:41, 2014.
- [10] B. Dhingra, J. Cole, J. M. Eisenschlos, D. Gillick, J. Eisenstein, and W. W. Cohen, “Time-Aware Language Models as Temporal Knowledge Bases,” *Trans. Assoc. Comput. Linguist.*, vol. 10, pp. 257–273, 2022.
- [11] D. Huwiler, K. Stockinger, and J. Fürst, “VersionRAG: Version-Aware Retrieval-Augmented Generation for Evolving Documents,” arXiv:2510.08109, 2025.
- [12] Y. Zhao et al., “TimelyRAG: Semantic-Temporal Hybrid Retrieval for Time-Critical Question Answering in Overlapping-Evolving Documents,” arXiv:2609.13524, 2026.
- [13] Y. A. Malkov and D. A. Yashunin, “Efficient and Robust Approximate Nearest Neighbor Search Using Hierarchical Navigable Small World Graphs,” *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 42, no. 4, pp. 824–836, 2020.
- [14] J. Johnson, M. Douze, and H. Jégou, “Billion-Scale Similarity Search with GPUs,” *IEEE Trans. Big Data*, vol. 7, no. 3, pp. 535–547, 2021.
- [15] C. Guo, G. Pleiss, Y. Sun, and K. Q. Weinberger, “On Calibration of Modern Neural Networks,” in *Proc. ICML, PMLR*, vol. 70, pp. 1321–1330, 2017.
- [16] P. He, X. Liu, J. Gao, and W. Chen, “DeBERTa: Decoding-Enhanced BERT with Disentangled Attention,” in *Int. Conf. Learning Representations*, 2021.
- [17] A. Asai, Z. Wu, Y. Wang, A. Sil, and H. Hajishirzi, “Self-RAG: Learning to Retrieve, Generate, and Critique through Self-Reflection,” in *Int. Conf. Learning Representations*, 2024.
- [18] S. Es, J. James, L. Espinosa-Anke, and S. Schockaert, “RAGAs: Automated Evaluation of Retrieval Augmented Generation,” in *Proc. EACL: System Demonstrations*, pp. 150–158, 2024.
- [19] J. Saad-Falcon, O. Khattab, C. Potts, and M. Zaharia, “ARES: An Automated Evaluation Framework for Retrieval-Augmented Generation Systems,” in *Proc. NAACL-HLT*, pp. 338–354, 2024.
- [20] C. Niu et al., “RAGTruth: A Hallucination Corpus for Developing Trustworthy Retrieval-Augmented Language Models,” in *Proc. ACL*, pp. 10862–10878, 2024.
- [21] A. Al Masoud, M. Arazzi, and A. Nocera, “SD-RAG: A Prompt-Injection-Resilient Framework for Selective Disclosure in Retrieval-Augmented Generation,” arXiv:2601.11199, 2026.
- [22] T. Thuillier, L. Lavaur, J. François, R. State, and W. Ferguson, “Key-Parametrized Embeddings for Access Control in Retrieval-Augmented Generation,” in *Proc. IEEE 50th Ann. Computers, Software, and Applications Conf.*, pp. 2581–2586, 2026.
- [23] U. R. Saxena, S. Shailendra, R. Kadel, M. T. Islam, and A. Sharma, “Privacy-provisioned-permission-aware RAG in cloud using identity and access management, trust and large-language models,” *Sci. Rep.*, vol. 16, Art. 71936, 2026.
- [24] Y. Mu et al., “Towards Secure Retrieval-Augmented Generation: A Comprehensive Review of Threats, Defenses and Benchmarks,” arXiv:2603.21654, 2026.
- [25] B. Palanisamy, G. S. S. Chalapathi, V. Hassija, and R. Buyya, “Security and Privacy in Retrieval-Augmented Generation: Architectures, Threats, Defenses, and Future Directions for Building Trustworthy Systems,” arXiv:2606.25533, 2026.

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